Requirements & Design Specification

Project: KETCo Unified Digital Platform
Organization: The Kerry Environmental Testing Cooperative (Tralee, Co. Kerry)
Date: October 2026
Version: 1.0.0

1. Problem Definition & Scope

Organization Profile

The Kerry Environmental Testing Cooperative (KETCo) is an environmental testing service provider based in Tralee, County Kerry. The cooperative provides water testing services and soil testing services for rural and local stakeholders. Customers can deliver samples directly to the central laboratory in Tralee. Alternatively, field staff can collect samples at scheduled collection points in rural areas.

Testing requests have three main causes:

- General environmental monitoring
- Specific project work
- Targeted investigations after an unexpected environmental result

Present Process

Intake & Metadata Logging: Lab staff write customer requests, sample labels, collection data, and test assignments manually. Staff use paper submission forms, local spreadsheets, emails, and physical lab worksheets.

Sample Harvesting: Customers, dedicated field technicians, or regional partner organizations collect samples in the field.

Receipt & Quality Screening: Lab staff manually examine the quality of incoming samples. Staff check for leaks, correct labels, temperature changes, or low volume. Then, staff write a manual laboratory reference number on the sample.

Internal Logistical Routing: Samples move between reception, preparation, testing, and temporary storage stations. Staff track these physical steps with paper bench sheets and handwritten labels on the bottles.

Testing & Referral Submissions: The laboratory manager distributes tests to analysts based on machine availability and staff training. Analysts frequently split samples into smaller fractions or send samples to external specialist laboratories.

Review & Reporting: Analysts record raw measurements and do manual calculations. An authorized reviewer must complete a technical review of the data before the laboratory issues an approved report. If

staff find errors after report release, staff must do a manual correction workflow to create a replacement report.

Core Business Problems & Consequences

Administrative Resource Drain: Staff write identical tracking data many times in separate paper logs and spreadsheets. This manual work creates large operational bottlenecks.

Data Ambiguity & Entry Errors: Staff use inconsistent names for identical geographic locations. Handwritten bottle labels are difficult to read, which makes accurate intake screening difficult.

Critical Custody Failure: Staff use fragmented paper bench sheets, which makes a complete, audit-ready custody history difficult to make. Lab staff mix up similar sample bottles in the same batch.

Quality Control & Sample Loss Risk: The laboratory does not have a single method to verify sample container types, preservation methods, and time limits. Analysts test degraded samples, which creates invalid data.

Information Security & Reporting Vulnerabilities: Unverified provisional values can be shown to clients before technical review. The system does not isolate customer data, which risks exposing commercially sensitive business results to unrelated customers.

Desired Outcomes

Unified Digital Platform: Replace separate paper, spreadsheet, and email silos with one central computer system.

Digital Chain of Custody: Create an unalterable digital log that tracks samples from collection to disposal to prevent bottle mix-ups.

Geographical Standardization: Use strict database boundaries to separate broad customer location descriptions from precise GPS sampling coordinates.

Real-Time Operational Visibility: Provide live data to the Laboratory Manager about samples that wait for checks, active tests, instrument delays, and reports that wait for approval.

Versioned Report Governance: Send secure reports that keep an unalterable history of report versions, specific reasons for revision, and recipient logs.

Known Evidence

- The system must manage only water and soil sample types.
- Internet access is stable at the Tralee office but unreliable at rural collection points.
- One request can have many samples, and one sample can have many tests.
- Database records for internal sample fractions must link to the original parent sample reference.
- Measurement instruments require calibration and quality control (QC) checks. Check failures delay scheduled testing.
- Unusual results are data anomalies. They do not automatically prove a pollution incident or confirm an environmental cause.

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Boundary & Scope

In-Scope Functions:

- **Request & Suitability Intake:** Record customer requests digitally, track sample suitability metrics automatically, and track provisional acceptance.
- **Spatial Coordination Mapping:** Separate broad sampling locations from precise GPS coordinates in the database.
- **Digital Chain of Custody Log:** Create continuous, timestamped tracking of internal sample movement. Record handler IDs, station changes, and bench sheet references.
- **Parent-Child Relationship Architecture:** Map data to link original environmental samples to internal split fractions and external laboratory referral records.
- **Workflow & Allocation Management:** Match tests to analysts based on training certificates. Track live test statuses and record raw measurements with provisional calculations.
- **Instrument Performance Tracking:** Log equipment availability statuses, routine calibration results, and pre-test QC checks.
- **Versioned Reporting Engine:** Generate final reports with chronological version numbers, mandatory revision notes, and secure contact access control logs.

Out-of-Scope Functions:

- **Automated Instrument Interfacing:** The system will not connect directly to laboratory hardware with APIs or drivers. Analysts must manually input all machine data, raw observations, and measurements.
- **Regulatory Enforcement Tracking:** The system will not evaluate statutory limits or send automated compliance alerts to government authorities.
- **Financial Accounting & Billing:** The system will not process customer invoices, billing models, credit cards, or general ledger workflows.
- **Professional Consultation Delivery:** The system will not generate automated environmental conclusions or give professional environmental advice.

External Dependencies

- **Rural Network Infrastructure:** Mobile operations in the field depend on local mobile networks or offline data caching because rural internet is poor.
- **Partner Logistics Accuracy:** Early sample tracking depends on partner organizations that record field metadata correctly at rural drop-off points.
- **Third-Party Lab Lead Times:** Final reporting workflows depend on the processing speed and delivery schedules of external specialist laboratories.

Critical Information Gaps & Logical Uncertainties

- **Provisional Boundary Rules:** The specific criteria for provisional acceptance and the maximum holding duration before system rejection.

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- **Suitability Rejection Matrix:** A rule set that maps physical defects (like temperature thresholds or leaks) to automatic test unsuitability.
- **Sub-Sample Tracking Standards:** The database naming standard that links internal split fractions and external referral samples to the primary reference.
- **Authorization Privilege Matrix:** Definitions of user roles to show who can approve a method departure, override a failed QC check, or start a manual re-test.
- **Dynamic SLA Logic:** System logic to recalculate estimated completion dates when instruments stop operating.
- **Correction Approval Governance:** The review workflow and security clearances necessary to approve and distribute replacement reports.
- **Data Isolation Access Controls:** An access matrix to restrict view permissions by customer type, field technician, analyst, and external partner.

2. Stakeholders, Actors, & Goals

Stakeholder Catalogue

Direct Stakeholders (Actors):

- **Reception Technician:** Generates unique laboratory references, checks paperwork, records receipt variables (temperature, volume, leaks), and manages 'Provisional Acceptance' states.
- **Field Technician:** Records precise GPS coordinates for sampling points, records early preservation actions, and starts the chain of custody log with offline capabilities.
- **Analyst:** Selects testing tasks based on certification, records raw measurements, records instrument calibrations, and flags unusual provisional values.
- **Quality Manager:** Audits provisional test results, verifies instrument calibration states, approves final laboratory reports, and authorizes version-controlled corrected reports.
- **Laboratory Manager:** Monitors operational dashboards for workflow bottlenecks, monitors instrument uptime, manages external referrals, and tracks unapproved corrected reports.
- **Customer / Community Group Nominated Contacts:** Submits online sample requests, tracks sample suitability status, receives alerts about delayed tests, and views final laboratory reports securely.

Indirect Stakeholders:

- **Community Groups & Environmental Project Sponsors:** Use KETCo testing data to get funding, present findings to local assemblies, or monitor community water sources.
- **Farms and Small Business Owners:** Require commercial privacy and accurate reports to make operational decisions without data exposure to competitors.
- **Cooperative Board Members / Strategic Leadership:** Require a system that removes manual data re-entry, protects KETCo from liability, and allows operational scaling.

External Stakeholders:

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- **Partner Organisation:** Coordinates rural collection logistics, delivers samples with metadata, and uses physical or digital interfaces at drop-off points.
- **External Specialist Laboratory:** Receives preserved sample fractions with parent references, performs specialized assays, and returns structured test results.
- **Environmental Regulatory & Accreditation Frameworks:** Mandate rules that separate raw instrument data, provisional calculations, and authorized results.

Actor-Goal Catalogue

Actor Name	Actor Type	System Goal / Core Concern	Influence Level
Customer / Community Contact	Primary	Submit requests, view suitability alerts, download authorized reports.	High
Reception Technician	Primary	Generate reference IDs, document sample conditions, manage provisional states.	High
Field Technician	Primary	Log GPS coordinates, start custody logs from rural zones.	Medium
Analyst	Primary	Run calibrations, log raw measurements, calculate provisional values.	High
Authorized Reviewer / Quality Manager	Primary	Verify test results against QC checks, authorize version-controlled reports.	High
Laboratory Manager	Primary	Monitor dashboards for sample bottlenecks, instrument downtime, and approvals.	High
Partner Organisation	Supporting	Synchronize rural drop-off metadata, verify physical custody transfers.	Medium
External Specialist Laboratory	Supporting	Accept split sample portions, return technical test results.	Medium
Cooperative Board / Leadership	Offstage	Ensure data integrity, prevent mix-ups, remove manual re-entry.	High
Compliance / ISO 17025 Auditor	Offstage	Verify immutable audit trails that separate raw data from final results.	High

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3. Requirements & Business Rules

Requirements Catalogue

ID	Subsystem / Type	Name	Description	Priority
FR-101	Intake & Sample Registration	Unique Identification Generation	The system must automatically generate and assign a unique laboratory_reference ID to every sample at the moment of receipt check.	Must
FR-102	Intake & Sample Registration	Spatial Coordinate Mapping	The system must record exact sampling_point_coordinates (GPS data) and enforce a link to a parent location_id.	Must
FR-103	Intake & Sample Registration	Conditional Acceptance Status Management	The system must let a reception technician assign a status of Provisionally Accepted and force the entry of receipt_condition_notes if data is incomplete.	Must
FR-104	Intake & Sample Registration	Sample Suitability Evaluation Matrix	The system must evaluate receipt_temperature, container_type_used, and elapsed time against test_type_code rules to verify if a sample is fit for specific assays.	Should
FR-105	Core Tracking & Custody	Parent-Child Fraction Splits	The system must support splitting a sample into fractions by creating a new sample record with a parent_sample_reference foreign key.	Must
FR-106	Core Tracking & Custody	Immutable Chain of Custody Logging	The system must automatically add a row to the Chain of Custody Log Entry table when a sample's transferred_to_location or handler_staff_id changes.	Must

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FR-107	Lab Operations & QC	Data Segregation & Promotion Workflow	The system must separate observation_measurements_raw and provisional_value fields from the reportable_result field until technical review is complete.	Must
FR-108	Lab Operations & QC	Dynamic Assignment Validation Check	The system must verify the assigned_analyst_id against the employee's certified_methods list before saving a test assignment.	Should
FR-109	Lab Operations & QC	Instrument Interoperability Block	The system must change a Requested Test Assignment status to Delayed if the linked instrument_id has a status of Out of Service or a QC check of Fail.	Must
FR-110	Outputs & Reports	Versioned Correction Tracking	The system must increment the report_version_number and force the entry of text into revision_reason_notes when a replacement report is generated.	Must
FR-111	Outputs & Reports	Management Alert Escalation Dashboard	The system must create a view that shows unapproved corrected reports, active is_unusual_result flags, external referrals, and instrument delays for the Laboratory Manager.	Should
NFR-201	Offline Queue Data Resilience	Capacity & Sync Reliability	The field module must store up to 200 data logs offline when network connection drops. It must automatically synchronize data with the central Tralee database within 30 seconds of network reconnection.	Must

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NFR-202	Multi-Tenant Tenant Separation	Security / Data Privacy	The data tier must filter database queries using the commercial_sensitivity_flag and customer_id. The system must return an HTTP 403 Forbidden error if users try to read data outside their authorized scope.	Must
NFR-203	Automated Alert Notification Latency	Performance / Speed	The notification subsystem must send automated email or SMS alerts to the nomination_contacts_list within 5 minutes when a sample changes to a Rejected or Delayed status.	Should
NFR-204	Out-of-Scope System Boundaries	Maintainability / Boundary Constraint	The system architecture must have zero software dependencies on external modules for client invoicing, automated sensor telemetry parsing, or regulatory enforcement actions.	Must

Business Rules & Constraints Register

Role-Based Testing Assignments: Staff can assign tests to analysts only if the analyst has recorded training records for that specific method.

Mandatory Data Separation: The system must separate general geographic descriptions from precise GPS data in the database.

Result Verification and Release Hierarchy: Raw data from an instrument are provisional values. The system must prevent the release of provisional values to customers until a Quality Manager approves the final report.

Data Access Restrictions: If a customer record has a commercial sensitivity flag, the system must hide these testing results from all other customers.

Assumptions & Constraints

Provisional Sample Lifespan: The system assumes a maximum holding shelf-life of 48 hours for provisional samples before automated rejection occurs.

Component-Level Suitability: The system assumes that if one test fails sample suitability, other compliant tests on the same sample can stay active.

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Chain of Custody Inheritance: Sample portions or external referrals automatically inherit the primary Laboratory Reference ID as a parent reference.

Unreliable Rural Connectivity: Internet access is stable at the Tralee office but unreliable at rural collection points. The field module must support offline data entry.

Strict Out-of-Scope Boundaries: The project must exclude instrument hardware APIs, compliance alerts, customer invoicing, and professional advice.

Risks

Bottle Confusion & Chain Traceability Breakdowns: If the digital custody process is too complex, lab staff might bypass it, which will continue sample mix-ups.

Scope Creep from Premature Instrument Integration: Misaligning the boundary between manual entry and machine data extraction will cause project delays.

Data Siloing Between Roles: Failing to reconcile what data each role is authorized to see could cause data leaks or visibility blocks.

Clarification Questions

- **Provisional Windows:** What is the maximum time a sample can stay provisionally accepted before automatic rejection?
- **Suitability Matrix:** Is there a matrix that maps specific container defects directly to the tests they invalidate?
- **Approval Authorities:** Which specific roles can authorize repeat tests, method departures, or corrected reports?
- **Customer Communication Protocols:** Should updated completion dates notify the customer immediately or wait for manager review?
- **Granular Access Control:** What specific fields should be visible to partner organizations versus field technicians?

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4. Behavioural & Use-Case Modelling

UC-101: Register Sample & Verify Suitability

ID / Name	UC-101 Register Sample & Verify Suitability
Actor	Reception Technician
Preconditions	1. System is online. 2. Physical sample arrives at Tralee lab. 3. Sample Request exists.
Trigger	Technician starts intake on the system.
Basic Flow	1. Select request_id. 2. Select Sampling Location and precise GPS Sampling Point. 3. Input: sample_type, collected_date_time, collector_id, container_type_used, receipt_temperature, and receipt_quantity_volume. 4. System generates unique laboratory_reference ID. 5. System evaluates sample metrics against matrix rules. 6. System checks pass; sets acceptance_status to "Accepted". 7. System creates first Chain of Custody Log Entry (sets location to "Reception"). 8. System shows confirmation text.
Alternative Flows	**AF-1: Data Incomplete / Near Limits** • System prompts for receipt_condition_notes. • Technician inputs data defects. • System sets status to "Provisionally Accepted". • System starts 48-hour rejection timer. • Go to Step 7. **AF-2: Split Sample (Parent-Child)** • Operator selects split command. • System creates sub-sample record. • System copies parent ID to parent_sample_reference. • For external tests: Set is_external_referral to "True" and enter external_laboratory_id.
Exception Flows	**EF-1: Critical Defect (Leak, Expired Time, Wrong Container)** • System detects critical rule failure. • System sets status to "Rejected" and locks sample. • Subsystem sends alert to nomination_contacts_list within 5 minutes. • Process ends in failure.
Postconditions	**Success:** Sample saved with unique ID; status is "Accepted" or "Provisionally Accepted"; custody log starts. **Failure:** Status is "Rejected"; testing is blocked; customer is notified.

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Rules	• **GPS:** Separate broad locations from precise GPS coordinates. • **Timer:** Provisional state expires after 48 hours. • **Privacy:** If commercial_sensitivity_flag is "True", hide data from other customers.

UC-102: Track Field Collection & Map Coordinates

ID / Name	UC-102 Track Field Collection & Map Coordinates
Actor	Field Technician
Preconditions	1. Mobile application is open in offline mode or online mode. 2. Valid Sample Request exists.
Trigger	Technician extracts a physical sample at a rural point.
Basic Flow	1. Select active request_id. 2. Capture precise GPS sampling_point_coordinates. 3. Enforce link to parent broad location_id. 4. Log collected_date_time, preservation_methods_applied, and container_type_used. 5. Start initial Chain of Custody Log Entry locally. 6. Save transaction to local cache.
Alternative Flows	**AF-1: Network Connection Restored** • System detects active internet access at Tralee office. • System auto-triggers synchronization within 30 seconds. • Uploads up to 200 cached logs to central database.
Exception Flows	**EF-1: Invalid Spatial Data** • GPS signal fails or coordinate string is corrupt. • System blocks log submission and requests manual location checkpoint overrides.
Postconditions	**Success:** Precise coordinates mapped; offline log created and queued for synchronization.
Rules	• **Offline Capacity:** Store up to 200 data logs locally without server network drops. • **Isolation:** Strict database isolation prevents coordinate leaks to external entities.

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UC-103: Log Chain of Custody & Internal Transfers

ID / Name	UC-103 Log Chain of Custody & Internal Transfers
Actor	Reception Technician / Analyst
Preconditions	1. Sample has valid laboratory_reference ID. 2. Active custody log row exists.
Trigger	Physical bottle moves to a new internal station (e.g., preparation, testing, storage).
Basic Flow	1. Scan or input target laboratory_reference ID. 2. Select active transferred_to_location. 3. Input mandatory bench_sheet_reference. 4. System auto-captures current handler_staff_id and transfer_date_time. 5. Append new row entry to Chain of Custody Log Entry table.
Alternative Flows	**AF-1: Split Portion Inheritance** • At step 1: Sample is a split child portion. • System checks parent_sample_reference and copies historic custody metadata automatically.
Exception Flows	**EF-1: Incomplete Station Step** • Handler leaves station fields blank or types invalid handler ID. • System halts custody row generation; flags tracking error to prevent bottle confusion.
Postconditions	**Success:** Unalterable digital log appended; tracking history remains continuous and audit-ready.
Rules	• **Immutability:** Existing rows in the custody log cannot be edited or deleted by any user role.

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UC-104: Calibrate Instrument & Verify QC Checks

ID / Name	UC-104 Calibrate Instrument & Verify QC Checks
Actor	Analyst
Preconditions	1. Master profile instrument_id exists in the registry.
Trigger	Routine scheduled check or pre-test validation is required.
Basic Flow	1. Select target instrument_id. 2. Select check_type (Routine Calibration or Pre-test QC Check). 3. Input numeric qc_metric_measurements from device display manually. 4. System matches metrics against acceptable error ranges. 5. Metrics pass; system updates qc_check_result to "Pass" and availability_status to "Available".
Alternative Flows	**AF-1: QC Check Failure** • At step 5: Error margins exceed limits; system sets qc_check_result to "Fail". • System flags machine availability_status as "Out of Service".
Exception Flows	None. (This use case establishes operational reference logs).
Postconditions	**Success:** Instrument profile updated with new timestamped calibration log entry.
Rules	• **Interoperability Block:** No test assignments can execute on devices with a "Fail" status.

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UC-105: Execute Test Assignment & Log Measurements

ID / Name	UC-105 Execute Test Assignment & Log Measurements
Actor	Analyst
Preconditions	1. laboratory_reference is "Accepted". 2. Device calibration log reads "Pass".
Trigger	Workflow allocation pushes a pending Requested Test Assignment.
Basic Flow	1. Verify assigned_analyst_id aligns with certified_methods. 2. Change status to "In Progress". 3. Execute manual assay; input raw machine numbers into observation_measurements_raw. 4. Calculate and input the initial provisional_value. 5. Set assignment status to "Awaiting Review".
Alternative Flows	**AF-1: Instrument Delay Event** • System finds linked instrument is "Out of Service". • System auto-changes status to "Delayed" and triggers dynamic SLA recalculation. **AF-2: Method Departure Event** • Analyst alters path; system forces entry of data into analyst_notes_departures.
Exception Flows	**EF-1: Data Anomaly Detection** • Input numbers breach standard variance baselines. • System flags is_unusual_result as "True" and locks values for manager review.
Postconditions	**Success:** Raw and provisional data recorded; data fields are segregated from client view.
Rules	• **Data Segregation:** provisional_value field is completely hidden from customers.

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UC-106: Review Test Results & Authorize Report

ID / Name	UC-106 Review Test Results & Authorize Report
Actor	Quality Manager / Authorized Reviewer
Preconditions	1. All sample assignments show status "Awaiting Review".
Trigger	Manager opens the verification profile list.
Basic Flow	1. Select target request_id report structure. 2. Audit matching observation_measurements_raw and instrument calibration state histories. 3. Confirm data integrity; system copies provisional_value data into reportable_result. 4. Click authorization button; system creates a new Laboratory Report record. 5. Set report_version_number to 1 and report_type to "Approved Report". 6. Save authorised_reviewer_id and timestamp.
Alternative Flows	**AF-1: Re-Test / Repeat Verification Trigger** • Manager finds a calculation flaw or method variance error. • Manager populates repeat_testing_authorised_by field with their staff ID. • Re-sets test status back to "Scheduled"; clears raw value text buffers.
Exception Flows	None.
Postconditions	**Success:** Finalized results promoted to report status; report unlocked for customer portal access.
Rules	• **Hierarchy:** Only accounts matching the Quality Manager role can generate authorization keys.

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UC-107: Authorize Version-Controlled Corrected Report

ID / Name	UC-107 Authorize Version-Controlled Corrected Report
Actor	Quality Manager
Preconditions	1. An initial version-1 Approved Report was previously released.
Trigger	Staff discover an accurate post-release error requiring a manual data revision.
Basic Flow	1. Open the historically released report_id template record. 2. Modify incorrect numerical metrics inside the specific reportable_result field. 3. System forces entry into the revision_reason_notes text area. 4. Type specific reasons explaining the data shift. 5. System auto-increments report_version_number sequentially (e.g., from 1 to 2). 6. Set report_type flag parameter to "Corrected Report". 7. Append digital signature token, release files, and update recipient_delivery_log.
Alternative Flows	None.
Exception Flows	**EF-1: Blank Justification Error** • Manager attempts to save changes without completing explanation text blocks. • System blocks execution; report version roll-forward is cancelled.
Postconditions	**Success:** Correction generated; unalterable history logs containing past text variants are preserved.
Rules	• **Governance:** Corrected reports stay blocked from release until a final Quality Manager override.

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UC-108: Monitor Operations Dashboard & Bottlenecks

ID / Name	UC-108 Monitor Operations Dashboard & Bottlenecks
Actor	Laboratory Manager
Preconditions	1. Central system holds real-time transaction tables.
Trigger	Manager logs into system console to review operational workflow velocity.
Basic Flow	1. Open live operations management aggregation dashboard. 2. System queries database schema for active operations metrics. 3. Render summary view containing: • Count of samples awaiting intake verification check sequences. • Count of assignments running or marked as delayed. • Active is_unusual_result anomaly indicator flags. • Instrument down-time statuses and external referrals. • Pending unapproved versioned corrected reports.
Alternative Flows	None.
Exception Flows	None.
Postconditions	**Success:** Manager obtains visibility into performance bottlenecks; system data remains unedited.
Rules	• **Scope Limit:** View provides statistics only; it contains no invoicing metrics or automation APIs.

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UC-109: Access Authorized Laboratory Reports Securely

ID / Name	UC-109 Access Authorized Laboratory Reports Securely
Actor	Customer / Community Group Nominated Contact
Preconditions	1. User account profile has secure access credentials. 2. Target report status is authorized.
Trigger	Client logs into online web portal interface to download completed data outputs.
Basic Flow	1. Request file display list matching active customer_id context filter. 2. System evaluates commercial_sensitivity_flag state configuration parameters. 3. System runs tenant-separation rules to hide unrelated customer profiles. 4. Select desired report_id item row. 5. System displays authorized reportable_result vectors and full chronological version stacks. 6. System logs portal access transaction event within recipient_delivery_log array structures.
Alternative Flows	**AF-1: Community Contact Retrieval** • At step 1: Client belongs to a public group. • System scans nomination_contacts_list to confirm multi-user viewing authorization.
Exception Flows	**EF-1: Multi-Tenant Breach Rejection** • Target customer_id mismatches session token context. • System blocks query immediately; returns an HTTP 403 Forbidden error response code.
Postconditions	**Success:** Authorized results displayed; commercial privacy parameters remain intact.
Rules	• **Tenant Segregation:** Query algorithms must systematically block exposure of sensitive commercial details.

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Gherkin Acceptance Scenarios

Feature 1: Unique Identification Generation

```
Feature: Unique Identification Generation
  Scenario: Generate unique laboratory reference ID at receipt check
    Given a sample arrives at the reception station with complete paperwork
    When the reception technician completes the receipt check
    Then the system must generate a unique laboratory_reference ID
    And the system must assign this ID to the sample record
```

Feature 2: Data Segregation and Promotion Workflow

```
Feature: Data Segregation and Promotion Workflow
  Scenario: Prevent release of provisional values before technical review
    Given an analyst logs a provisional_value for a requested test
    When a customer tries to view the laboratory report
    Then the system must hide the provisional_value from the customer
```

5. Domain Analysis & Conceptual Model

Candidate-Concept Analysis

Candidate Term	Initial Classification	Decision	Justification & Evidence
Laboratory Reference ID	Attribute	Convert to Attribute	This is a unique identifier assigned to a sample record at intake.
Sample	Noun	Keep	This is a core entity that represents physical water or soil collected for testing.

Domain Glossary

Laboratory Reference ID: A unique identification code that the system automatically generates to track a single physical sample.

Provisional Value: A raw measurement or initial calculation that an analyst records before a formal technical review.

Conceptual Domain Model (UML)

(The PlantUML domain model reference: Linked to diagrams/domain-model.puml)

Modelling Rationale: Relationships ensure clear mapping where one request yields multiple samples, and one sample holds multiple test profiles. The parent-child layout guarantees complete sub-sample inheritance

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tracking without losing primary references.

6. Traceability & Project Execution

Traceability Matrix

Problem/Goal ID	Requirement ID	Use Case ID	Gherkin Scenario	Domain Concept
G-101	FR-101	UC-01	Scenario 1	Concept A

Recommended Repository Structure

```
ca1-analysis/
 ■■■ README.md
 ■■■ report/
 ■■■ requirements/
 ■■■ features/
 ■■■ diagrams/
 ■    ■■■ use-case-model.puml
 ■    ■■■ domain-model.puml
 ■■■ decisions/
 ■■■ references/
```

7. Reflection & AI Declaration

Critical Design Decisions & Revisions

Decision 1: Data Isolation Architecture
Enforced a hard separation tier inside database rules using customer IDs to fully isolate public community contact details from commercially sensitive private farming data, neutralizing corporate information leak hazards explicitly.

Decision 2: Manual Data Capture Block
Decoupled hardware sensor integrations to stay inside strict architectural bounds. Handlers manually upload instrumentation metrics to isolate system scope creep vulnerabilities.

Limitations & Unresolved Questions

- The explicit numeric values mapping specific container material types to specialized soil test boundaries need validation from final lab managers.
- Access permissions mapping out exactly what fields partner organizations can see during transit breaks down if metadata rules change.

AI Level 3 Disclosure & Interaction Log

AI Declaration: This specification was structured via generative engineering frameworks to map textual problem scopes into standard system analysis modules.

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Checked / Modified AI Suggestion:
AI Suggestion: Automatically link live testing equipment via wireless hardware APIs.
Action Taken: Rejected suggestion to remain strict with out-of-scope functional requirements and maintain system boundary definitions.

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